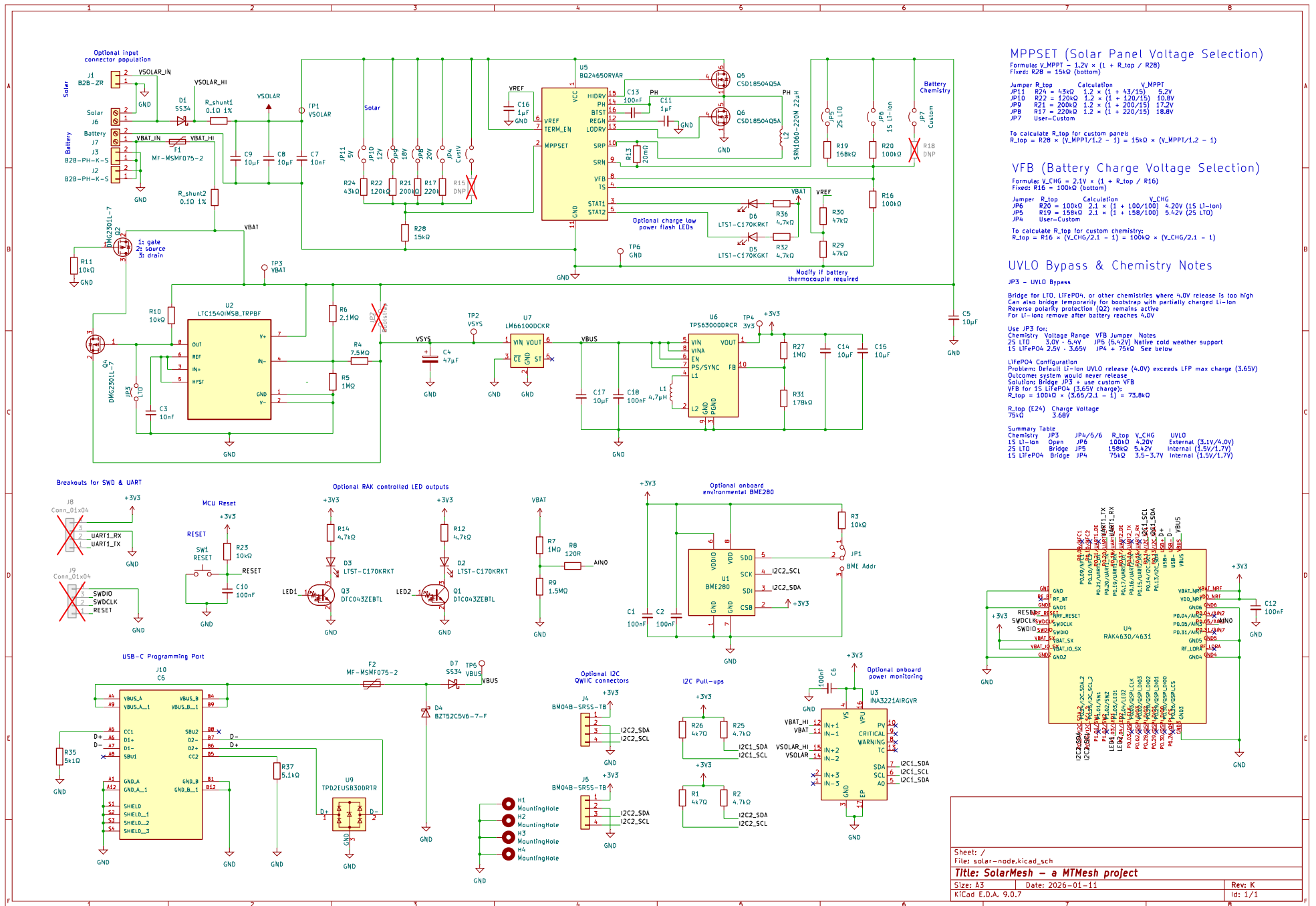




MPPSET (Solar Panel Voltage Selection)

Formula: $V_{MPP1} = 1.2V \times (1 + R_{top} / R28)$
 Fixed: $R28 = 15k\Omega$ (bottom)

Jumper R_top Calculation V_{MPP1}
 JP1: $R24 = 45k\Omega$ $1.2 \times (1 + 43/15) = 5.2V$
 JP10: $R22 = 120k\Omega$ $1.2 \times (1 + 120/15) = 10.8V$
 JP9: $R21 = 200k\Omega$ $1.2 \times (1 + 200/15) = 17.2V$
 JP8: $R17 = 220k\Omega$ $1.2 \times (1 + 220/15) = 18.8V$
 JP7: User-Custom

To calculate R_top for custom panels:
 $R_{top} = R28 \times (V_{MPP1}/1.2 - 1) = 15k\Omega \times (V_{MPP1}/1.2 - 1)$

VFB (Battery Charge Voltage Selection)

Formula: $V_{CHG} = 2.1V \times (1 + R_{top} / R16)$
 Fixed: $R16 = 100k\Omega$ (bottom)

Jumper R_top Calculation V_{CHG}
 JP6: $R20 = 100k\Omega$ $2.1 \times (1 + 100/100) = 4.20V$ (15 Li-Ion)
 JP5: $R19 = 158k\Omega$ $2.1 \times (1 + 158/100) = 5.42V$ (25 Li-Ion)
 JP4: User-Custom

To calculate R_top for custom chemistry:
 $R_{top} = R16 \times (V_{CHG}/2.1 - 1) = 100k\Omega \times (V_{CHG}/2.1 - 1)$

UVLO Bypass & Chemistry Notes

JP3 - UVLO Bypass

Bridge for LTO, LiFePO4, or other chemistries where 4.0V release is too high
 Can also bridge temporarily for bootstrap with partially charged Li-Ion
 Reverse polarity protection (Q2) remains active
 For Li-Ion: remove after battery reaches 4.0V

Use JP5 for:

Chemistry Voltage Range VFB Jumper Notes
 25 LTO 3.0V - 5.4V JP5 (5.4V) Native cold weather support
 15 LiFePO4 2.5V - 3.65V JP4 + 75kΩ See below

LiFePO4 Configuration

Problem: Default Li-Ion UVLO release (4.0V) exceeds LFP max charge (3.65V)

Outcomes: system would never release

Solution: Bridge JP3 - use custom VFB

VFB for 15 LiFePO4 (3.65V charge):

$R_{top} = 100k\Omega \times (3.65/2.1 - 1) = 73.8k\Omega$

R_top (E24) Charge Voltage

75kΩ 3.68V

Summary Table

Chemistry	JP3	JP4/5/6	R_top	V_CHG	UVLO External (3.1V/4.0V)
25 LTO	Open	JP6	100kΩ	4.20V	Internal (1.5V/1.7V)
15 LiFePO4	Bridge	JP5	158kΩ	5.42V	Internal (1.5V/1.7V)
15 LiFePO4	Bridge	JP4	75kΩ	3.5 - 3.7V	Internal (1.5V/1.7V)